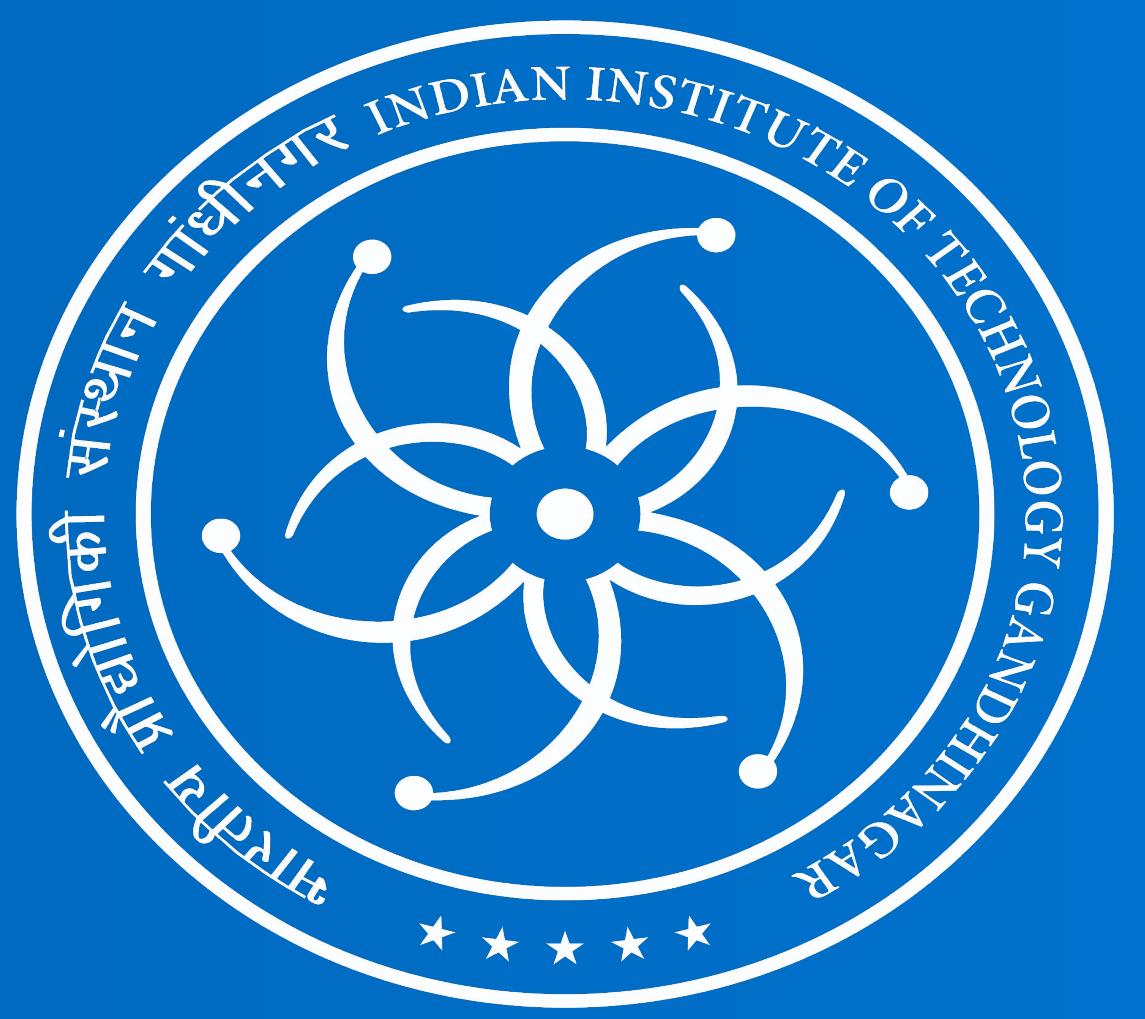


Integrated CFD-ML Framework for Simulation and Dynamic Mode Decomposition for Flow Over a Circular Cylinder



Dhvani Shah (24110325)
ME 299: Project Course
Project Instructor: Prof. Shubhangi Bansode

Objective

- Aim:** Develop a unified “end-to-end” Python workflow that triggers Ansys Fluent solvers, extracts data, and performs machine learning analysis using DMD validated against NASA experimental data.
- Base Case:** Automate steady and unsteady turbulent flow simulations over a Circular Cylinder utilizing Ansys PyFluent libraries.
- Analysis:** Apply Dynamic Mode Decomposition (DMD) to high-dimensional unsteady flow data to identify and analyze dominant flow structures and check for its efficiency.

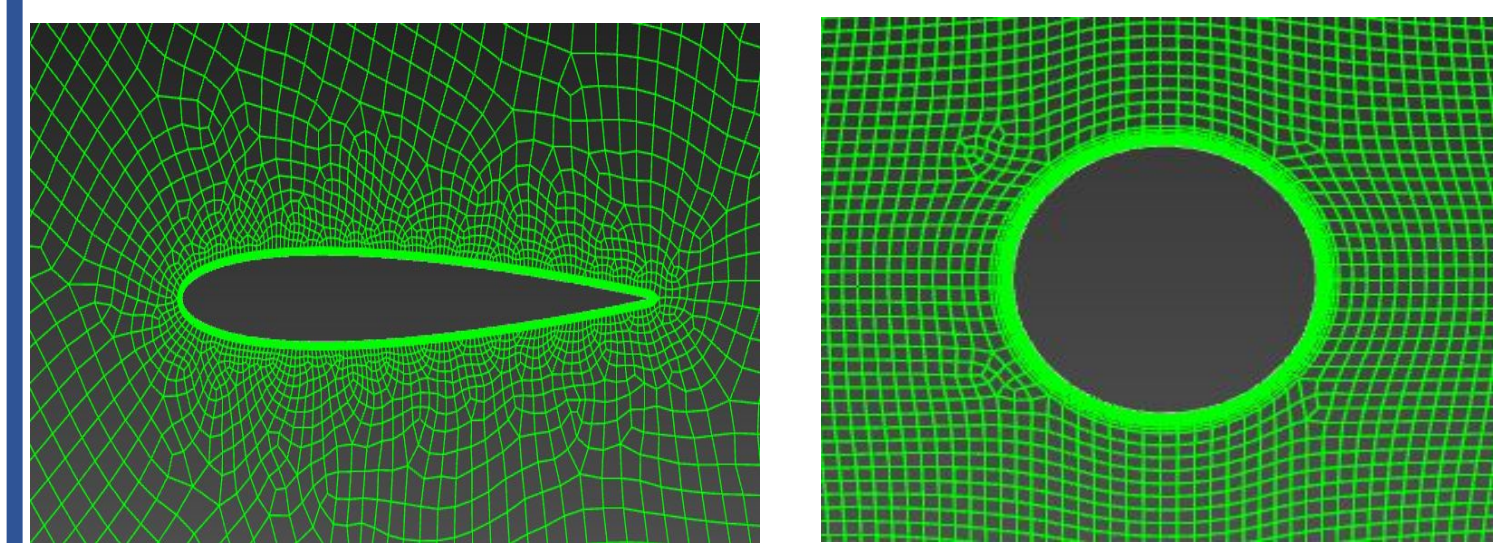
Methodology & Programmatic CFD Pipeline

1. Automated Workflow:

```
import ansys.fluent.core as pyFluent
import ansys.fluent.parametric as parametric
session = pyFluent.launch_fluent(no_gui=True)
```

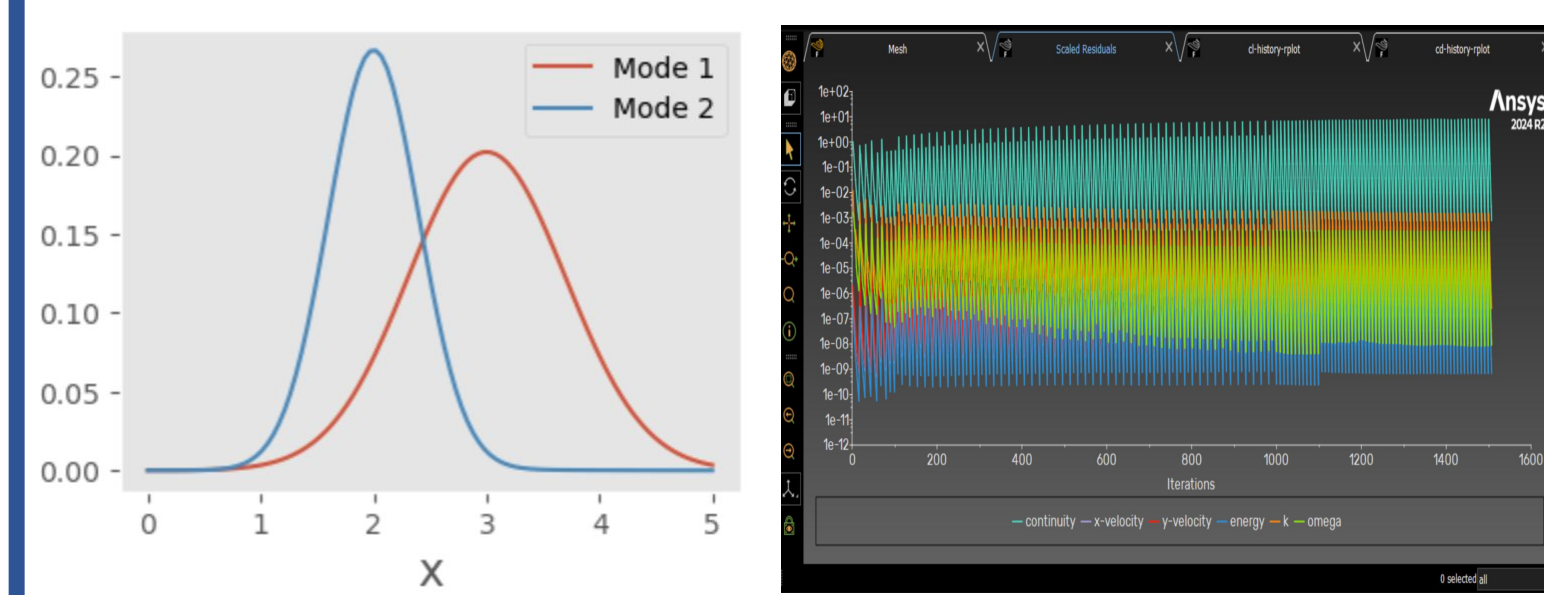
Implemented programmatic control over mesh handling, boundary conditions, and solver settings for transient conditions with reduced computational power.

2. Automated Geometry & Meshing Pipeline



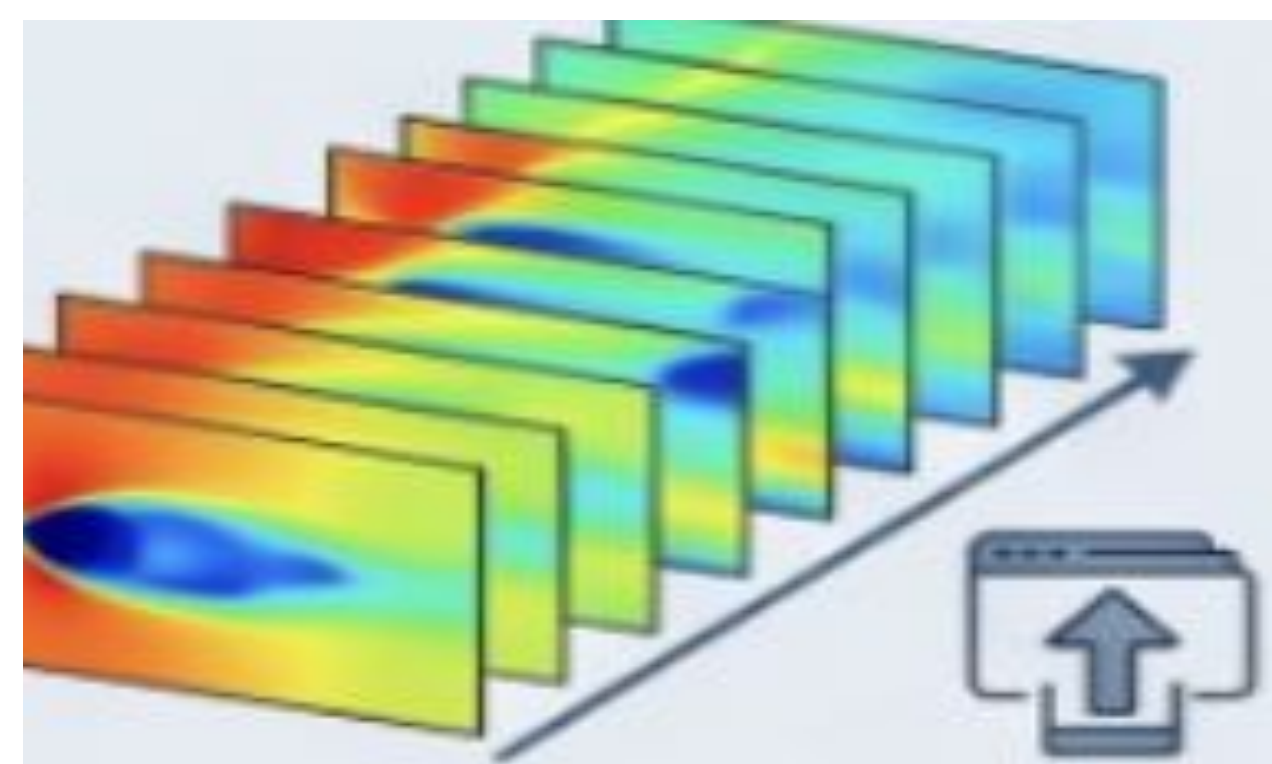
Generate geometry-adaptive Python scripts for diverse geometric topologies. Covers automated CAD processing, boundary layer (y+) insertion, volume volume meshing.

3. Transitional Solver Optimization & Setup



Determine optimal time-steps (Δt) from CFL, ensuring temporal accuracy and stability with k- ω SST turbulence model.

4. Data Acquisition



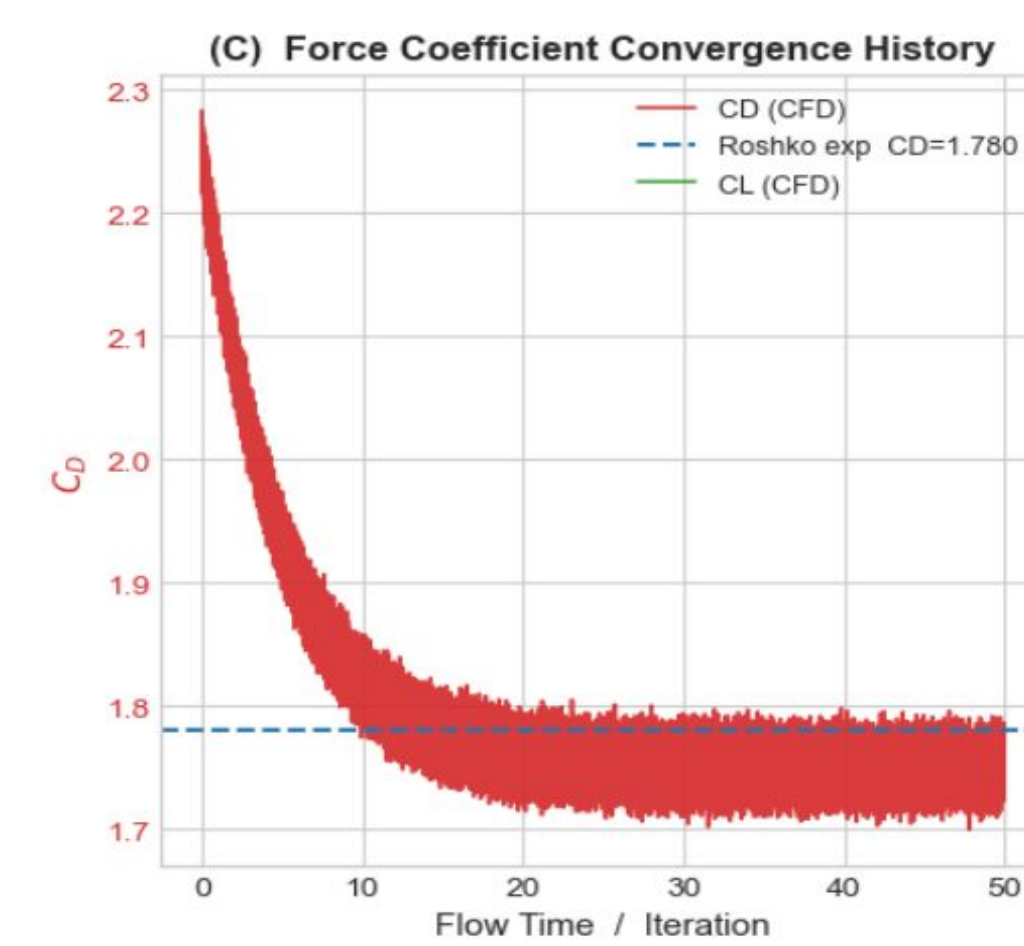
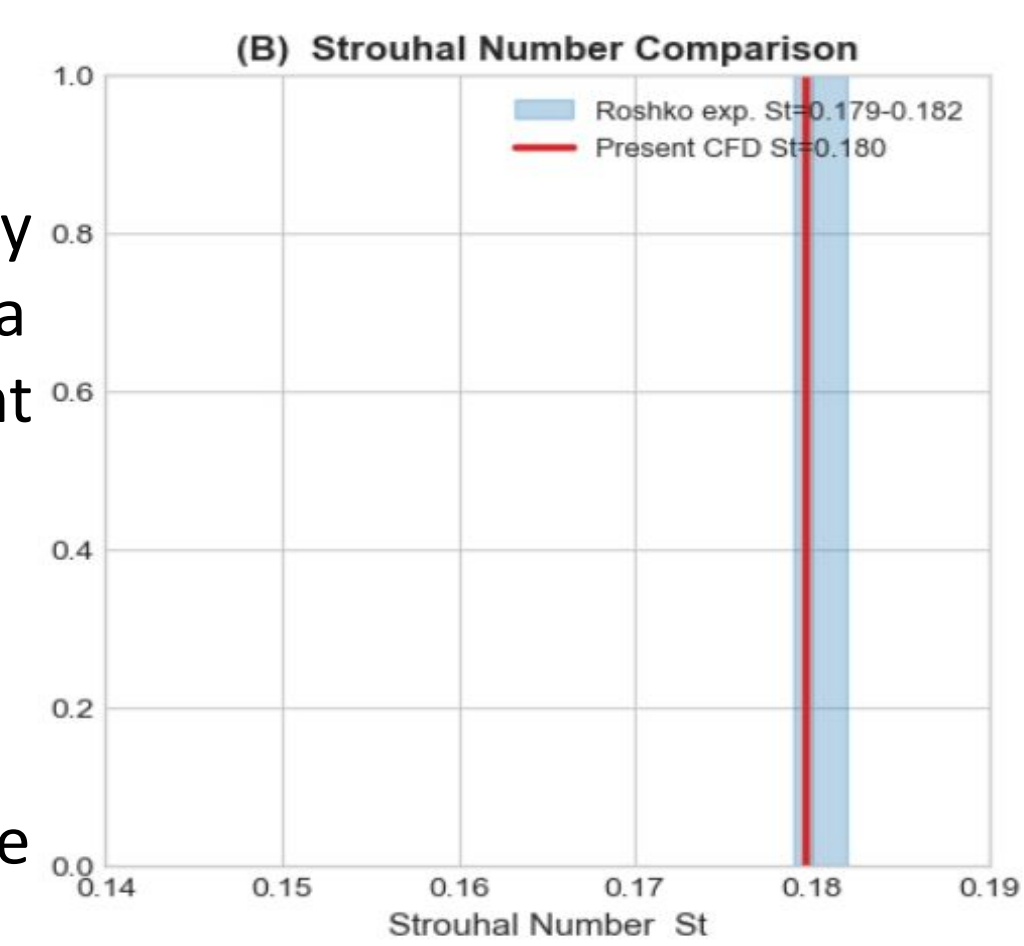
Automated capturing of time-resolved flow snapshots, replacing manual GUI-based methods

5. Baseline Validation

Cylinder flow successfully validated against NASA data within the PyFluent environment.

Drag coefficient error: -1.21%

Strouhal number error: -0.5%, confirming high accuracy of the CFD setup.



6. Dynamic Mode Decomposition (DMD) Framework (ROM Hub)

a. Data Feeding

Grid-Based Spatial/Temporal Extraction. Construct time shifted snapshot matrices X and X' for flow evolution.

b. Model Deployment

- SVD & DMD
- Execution & Mode Isolation
- Apply SVD on XXX to extract dominant modes.
- Retain only key modes ($r = 4$) capturing
- 99.9% kinetic energy.

c. Integrated Environment

- ML-Powered ROM Hub
- Build a Reduced Order Model (ROM) using extracted DMD modes and ML frameworks.
- Enable efficient reconstruction and prediction of flow dynamics.

d. Spatial Modes & Energy Representation

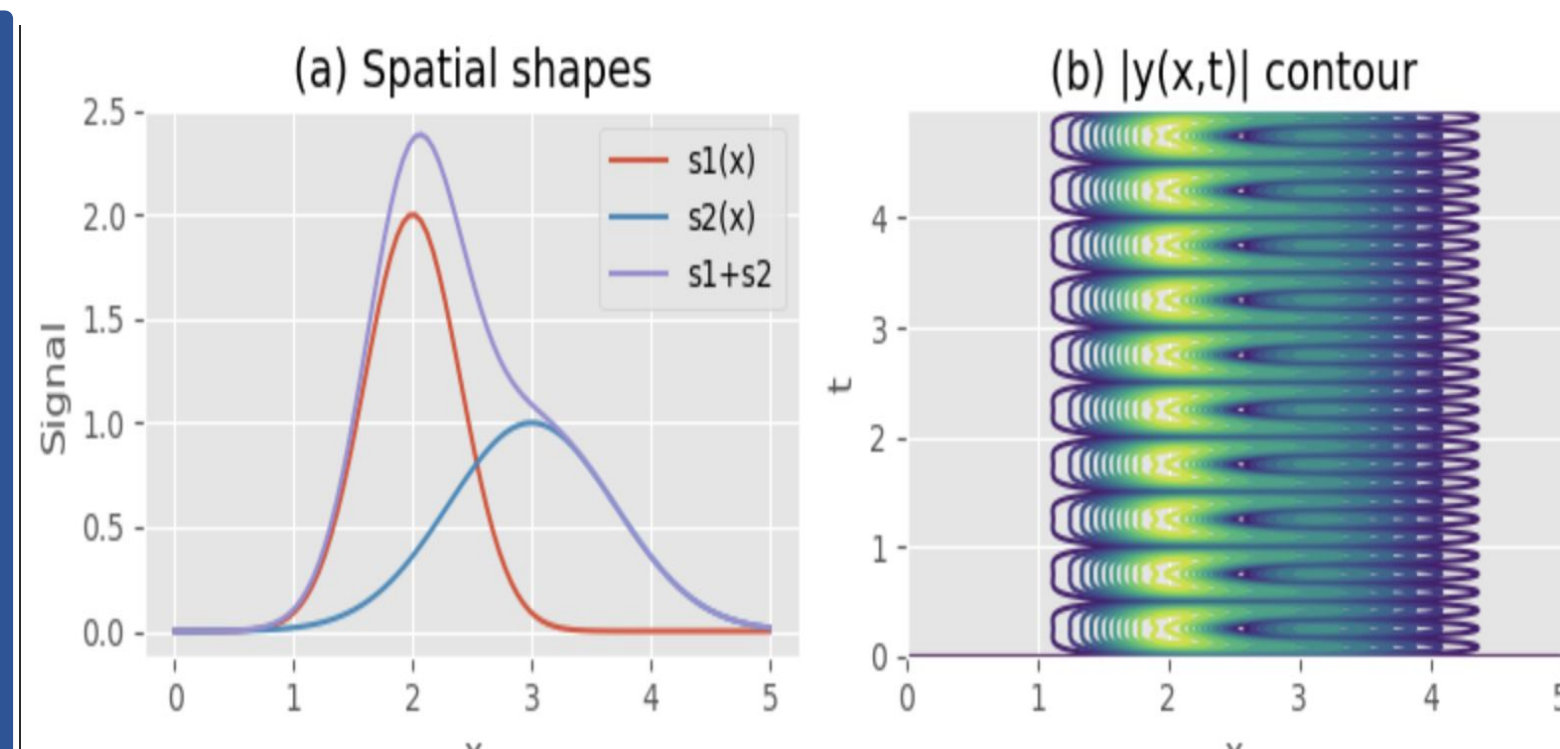
SVD extracts dominant spatial modes ϕ_k representing coherent flow structures in the field. The singular values σ_k quantify the energy contribution of each mode. A rapid decay in these values indicates that only a few modes are sufficient to capture the essential system dynamics.

f. DMD Temporal Evolution:

The system evolves as:

$$\mathbf{x}(t) = \sum_k \mathbf{a}_k(t) \phi_k$$

where each mode evolves dynamically using DMD eigenvalues.



e. Flow Field Reconstruction

The flow field is reconstructed as:

$$\mathbf{X} = \mathbf{U} \mathbf{\Sigma} \mathbf{V}^T$$

where: $\mathbf{U}$: spatial modes

$\mathbf{\Sigma}$: energy (singular values)

$\mathbf{V}$: temporal coefficients

$$\mathbf{X} = [\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_m]$$

$$\mathbf{X} = \mathbf{U} \mathbf{\Sigma} \mathbf{V}^T$$

$$\phi_k = \mathbf{U}_{:,k}$$

$$E_k = \frac{\sigma_k^2}{\sum_i \sigma_i^2}$$

$$\mathbf{x}(t) \approx \sum_{k=1}^r \mathbf{a}_k(t) \phi_k$$

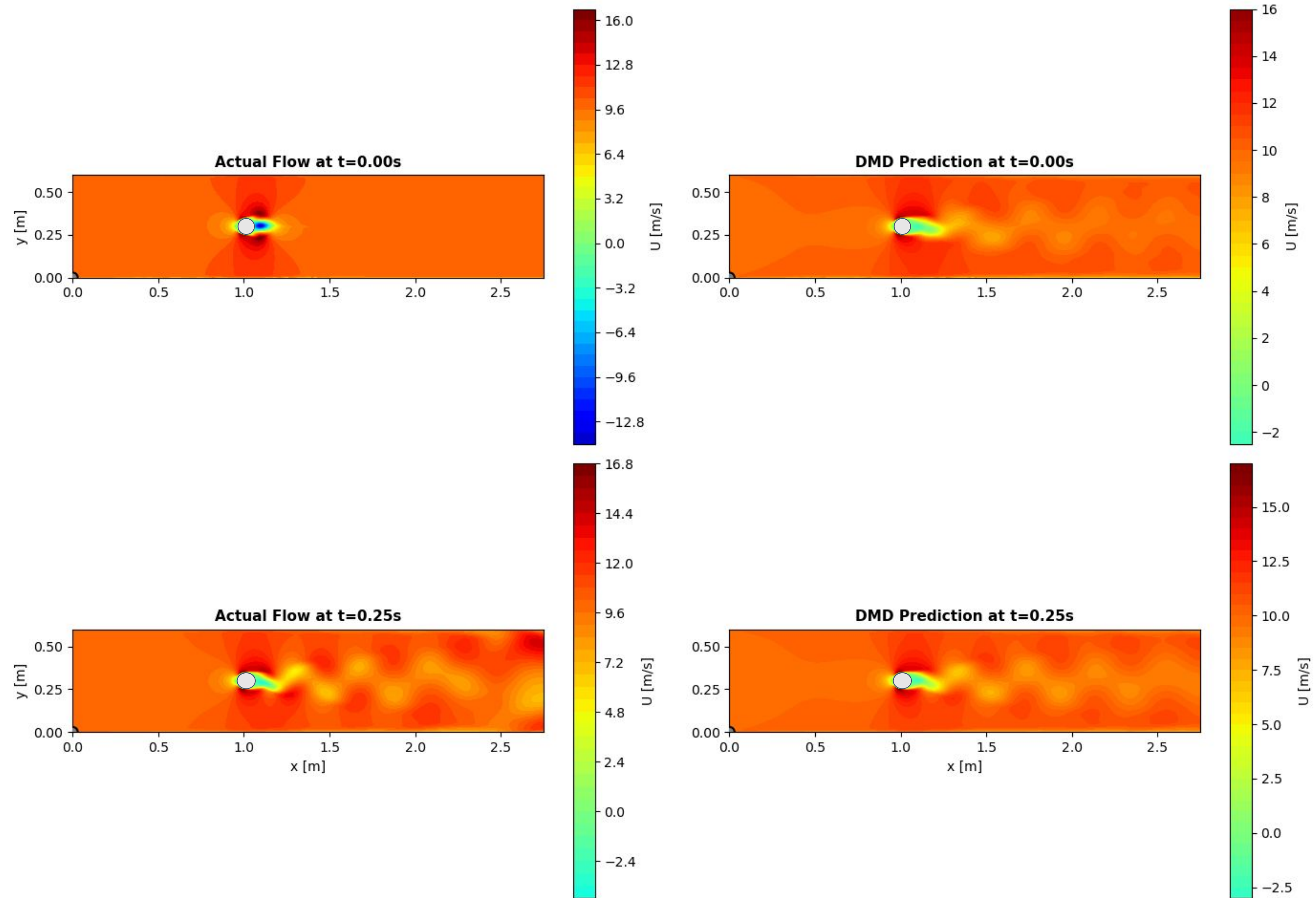
$$\mathbf{a}_k(t) = \sigma_k \mathbf{V}_{t,k}$$

$$\phi_k = \frac{1}{\sqrt{\lambda_k}} \mathbf{X} \mathbf{v}_k$$

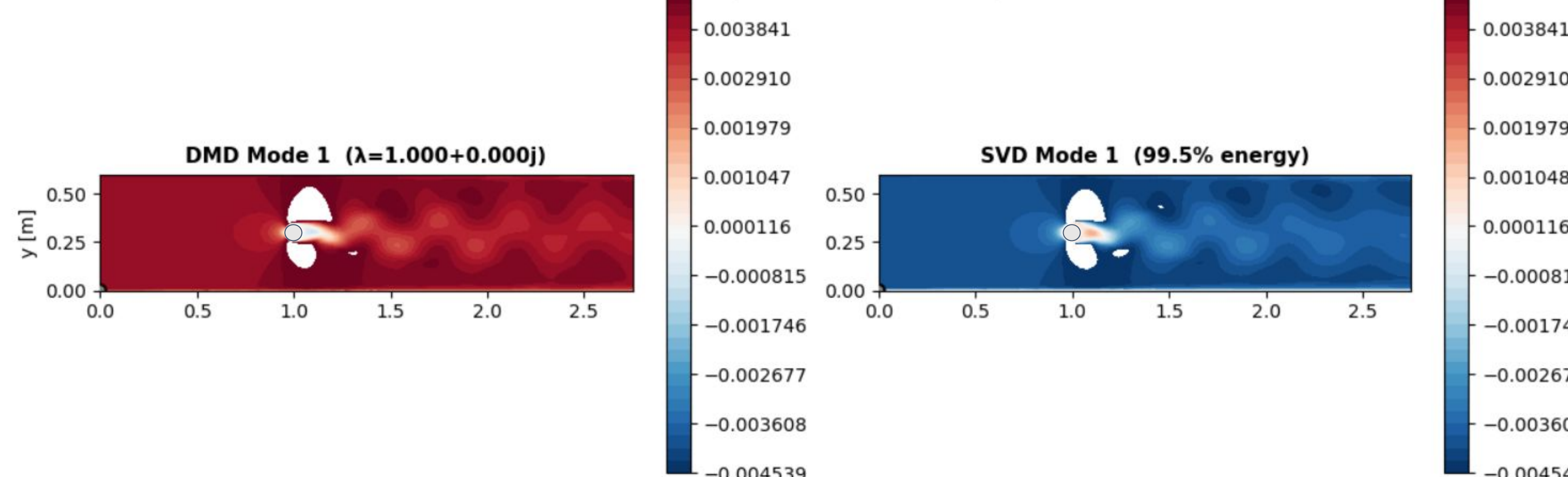
Results & Analysis

- DMD Reconstruction:** The DMD predictions show strong agreement with CFD snapshots across selected time instants, accurately capturing the dominant wake structures and flow evolution.
- Mode Consistency:** The leading DMD modes exhibit coherent vortex shedding patterns and closely match the corresponding SVD modes, confirming that key flow structures are well preserved.

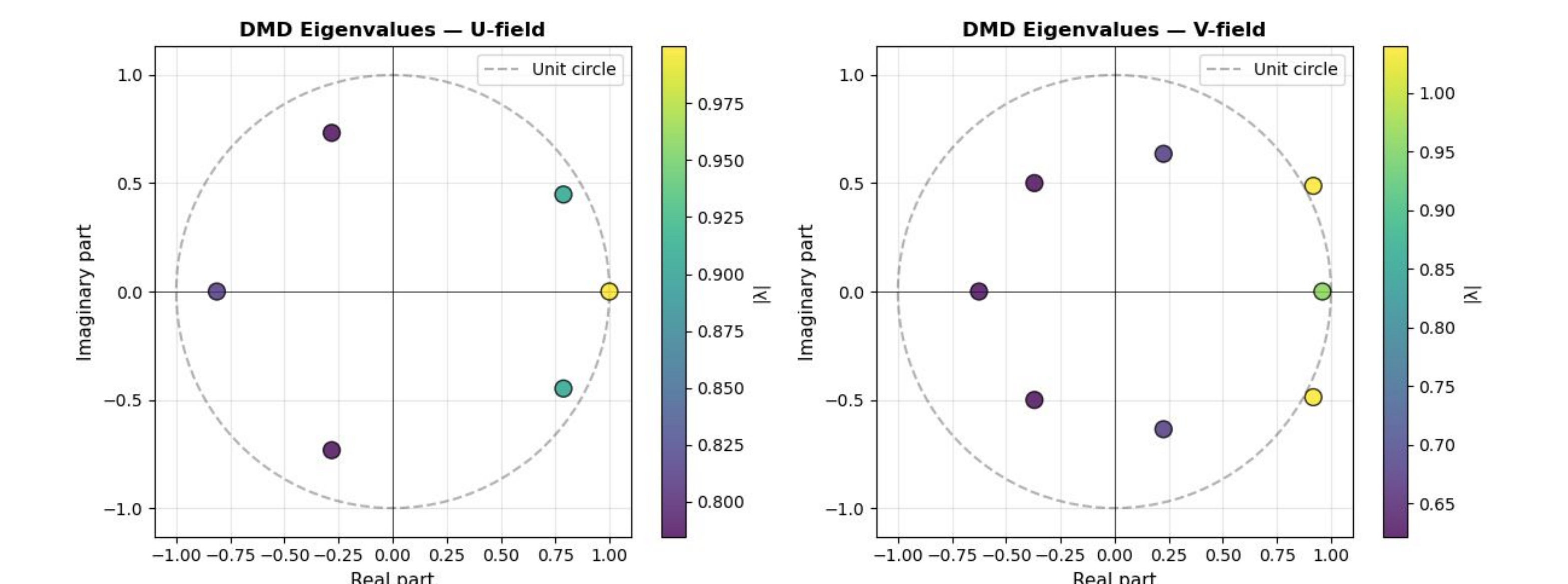
DMD Flow Prediction vs Actual



DMD vs SVD Spatial Modes Comparison



Inference



- Eigenvalue Analysis:** The DMD eigenvalues are distributed near the unit circle. These eigenvalues confirm that the flow is stable and oscillatory, and the DMD model accurately captures and reconstructs the dominant flow dynamics.
- By capturing dominant flow dynamics using DMD, we can replace expensive CFD simulations with faster predictive models, making aerodynamic analysis more efficient for practical engineering applications.

References

- C. J. Greenshields, Notes on Computational Fluid Dynamics: General Principles. Reading, UK: CFD Direct, 2022.
- Ladson, C. L., "Effects of Independent Variation of Mach and Reynolds Numbers on the Low-Speed Aerodynamic Characteristics of the NACA 0012 circular cylinder Section," NASA TM 4074, October 1988
- Ansys Innovation Resources